

## Analog CMOS IC Design (EE651)

### Assignment- 1

Total Marks – 10 points

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Date: 15/08/2025

Submission Date: 21/08/2025 (11:55 PM)

**Note:** Below is the list of questions on **MOS Transistors** and **Small-Signal Parameters of a MOS Transistor**. You are required to:

- Perform the simulations using **Cadence Virtuoso** with **SCL 180 nm PDK**.
- Prepare a detailed report based on your work.
- Submit the report for evaluation.

#### Report Guidelines:

- Include a **Conclusion** section at the end, briefly summarizing your insights and observations from the design process.
  - Name the report file as: RollNumber\_Assignment1.pdf
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1. Use n-type MOS and p-type MOS transistors of length 1000 nm and width 500nm. Plot transfer characteristics for  $V_{DS} = 100 \text{ mV}$  and  $V_{DS} = V_{DD}$  (1.8V) and output characteristics (for the VGS step 0.3 V). Explain your observations. [1pt]
2. Extract threshold voltage ( $V_{th}$ ) of MOS transistors. So, use **NMOS and PMOS** transistors of width 1000 nm and different lengths (200 nm, 400 nm, 600 nm, 800 nm and 1000 nm). Plot the following for the saturation region. [3pt]
  - a.  $V_{th}$  as a function of  $L_G$
  - b.  $I_{ds}$  as a function of  $L_G$
  - c.  $g_m$  as a function of  $L_G$
  - d.  $g_{ds}$  as a function of  $L_G$
  - e.  $\lambda$  as a function of  $L_G$

Explain your observations clearly.

3. Extract threshold voltage ( $V_{th}$ ),  $u_n C_{OX}$  of a MOS transistors. Use NMOS and PMOS transistors of length 200 nm and different widths (500 nm, 700 nm, 900 nm, 1100 nm and 1300 nm). Plot the following for the Linear and saturation regions. [2pt]
  - a.  $V_{th}$  as a function of Widths
  - b.  $I_{ds}$  as a function of Widths
  - c.  $g_m$  as a function of Widths
  - d.  $g_{ds}$  as a function of Widths

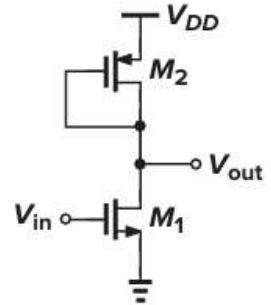
Explain your observations clearly.

4. Let us design and simulate the circuit shown below, using a diode-connected PMOS load. You will select the W/L ratios for both the NMOS and PMOS devices. Characterize the DC operating point, ensuring that the output voltage ( $V_{out}$ ) is 0.9 V and the power consumption does not exceed 30  $\mu\text{W}$ . [4pt]

**Tasks:**

- Determine and report the **DC operating point**, including  $I_D$ ,  $V_{GS}$ .
- Justify your choice of **device dimensions (W/L)** based on biasing and power constraints.
- Verify that the design meets the required  $V_{out}$  and power limit.
- Find the small signal gain, apply AC source at the input terminal and plot the output voltage.

Explain your observations clearly.



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Report should include,

- Transistor level circuit diagram (full system and individual sub-blocks) and sizing (with overdrive voltage) table of all transistors.
- Step-by-step design methodology/derivations/calculations (in brief).
- Overall power consumption and its block-wise breakup should be provided.
- Plots should be neatly labelled, and labels should be clearly visible
- Final results should be provided in tabular format
- Any other information as deemed necessary